6TH GRADE MATH · OPERATIONS OF FRACTIONS AND DECIMALS

Fraction Division: How Many Groups?

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Fraction Division: How Many Groups?** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Is $3 \div \frac{1}{2}$ more or less than 3? Explain.

   More: it is 6. Groups of one half fit twice into each whole.

2. In $2 \div \frac{3}{4}$, two full groups of $\frac{3}{4}$ fit and $\frac{1}{2}$ of a whole is left. Is the answer $2\frac{1}{2}$? Explain.

   No. The leftover $\frac{1}{2}$ is $\frac{2}{3}$ of a group of $\frac{3}{4}$, so the answer is $2\frac{2}{3}$.

2 Explore use the tool

1. Build $3 \div \frac{1}{2}$. How many groups fit? How many small pieces are in 3, and in one group?

   6 groups. 3 has 6 half-pieces and one group is 1 piece.

2. Build $2 \div \frac{3}{4}$. How many full groups fit, and what fraction of a group is left?

   2 full groups and $\frac{2}{3}$ of a group: $2\frac{2}{3}$.

3. Build $\frac{1}{2} \div \frac{3}{4}$. Why is the answer less than 1?

   $\frac{2}{3}$: the amount is smaller than one group, so less than one group fits.

4. Build $3 \div \frac{3}{2}$. Then compute $3 \times \frac{2}{3}$. What do you notice?

   Both are 2. Dividing by $\frac{3}{2}$ gives the same result as multiplying by $\frac{2}{3}$.

3 Record use the tool and fill in the table

Count the small pieces

Read the pieces from the sentence under the model.

Division	Pieces you have	Pieces in one group	Answer	Same as multiplying
$2 \div \frac{1}{4}$	8	1	8	$2 \times 4 = 8$
$3 \div \frac{3}{4}$	12	3	4	$3 \times \frac{4}{3} = 4$
$\frac{3}{4} \div \frac{1}{2}$	3	2	$1\frac{1}{2}$	$\frac{3}{4} \times 2 = \frac{3}{2}$
$4 \div \frac{5}{2}$	8	5	$1\frac{3}{5}$	$4 \times \frac{2}{5} = \frac{8}{5}$

4 Explain what did you discover?

1. a ÷ d asks: how many groups of size d fit into a?

2. When the divisor is less than 1, the quotient is greater than the dividend.

3. Dividing by $\frac{n}{m}$ gives the same answer as multiplying by $\frac{m}{n}$ (the reciprocal).

4. A leftover piece is a fraction of one group, not of one whole.

5. Use the bar model to explain why $2 \div \frac{3}{4}$ is $2\frac{2}{3}$ and not $2\frac{1}{2}$.

   Two groups of $\frac{3}{4}$ use $1\frac{1}{2}$, leaving $\frac{1}{2}$. One group is 3 quarters and the leftover is 2 quarters, so the leftover fills $\frac{2}{3}$ of a group.

5 Apply a new situation, no tool

1. How many $\frac{2}{3}$-cup servings are in 4 cups of rice?

   6 servings

2. A ribbon $\frac{7}{8}$ m long is cut into pieces $\frac{1}{4}$ m long. How many full pieces, and what fraction of a piece is left?

   3 full pieces and $\frac{1}{2}$ of a piece ($\frac{7}{8} \div \frac{1}{4} = 3\frac{1}{2}$).

3. Compute $1\frac{1}{2} \div \frac{3}{8}$.

   4

4. Without computing, is $5 \div \frac{9}{10}$ greater or less than 5? Why?

   Greater (it is $5\frac{5}{9}$), because the divisor is less than 1.

Teacher notes

Intended discovery: Division as measurement: count how many groups of the divisor fit. Cutting both amounts into equal pieces turns it into whole-number division and explains $a \div \frac{n}{m} = a \times \frac{m}{n}$.

Likely misconceptions:

- Naming the remainder as a fraction of the whole ($2\frac{1}{2}$) instead of a fraction of a group ($2\frac{2}{3}$).
- "Division always makes smaller" (Try this 2 asks for a counterexample).
- Flipping the dividend instead of the divisor.

Answers that can vary: Try this 1 and 5 accept many divisions (e.g., $3 \div \frac{1}{2}$ or $2 \div \frac{1}{3}$ for 6; $4 \div \frac{3}{2}$ for $2\frac{2}{3}$).